Titanic Survival Analysis

Understanding What Factors Determined Who Survived

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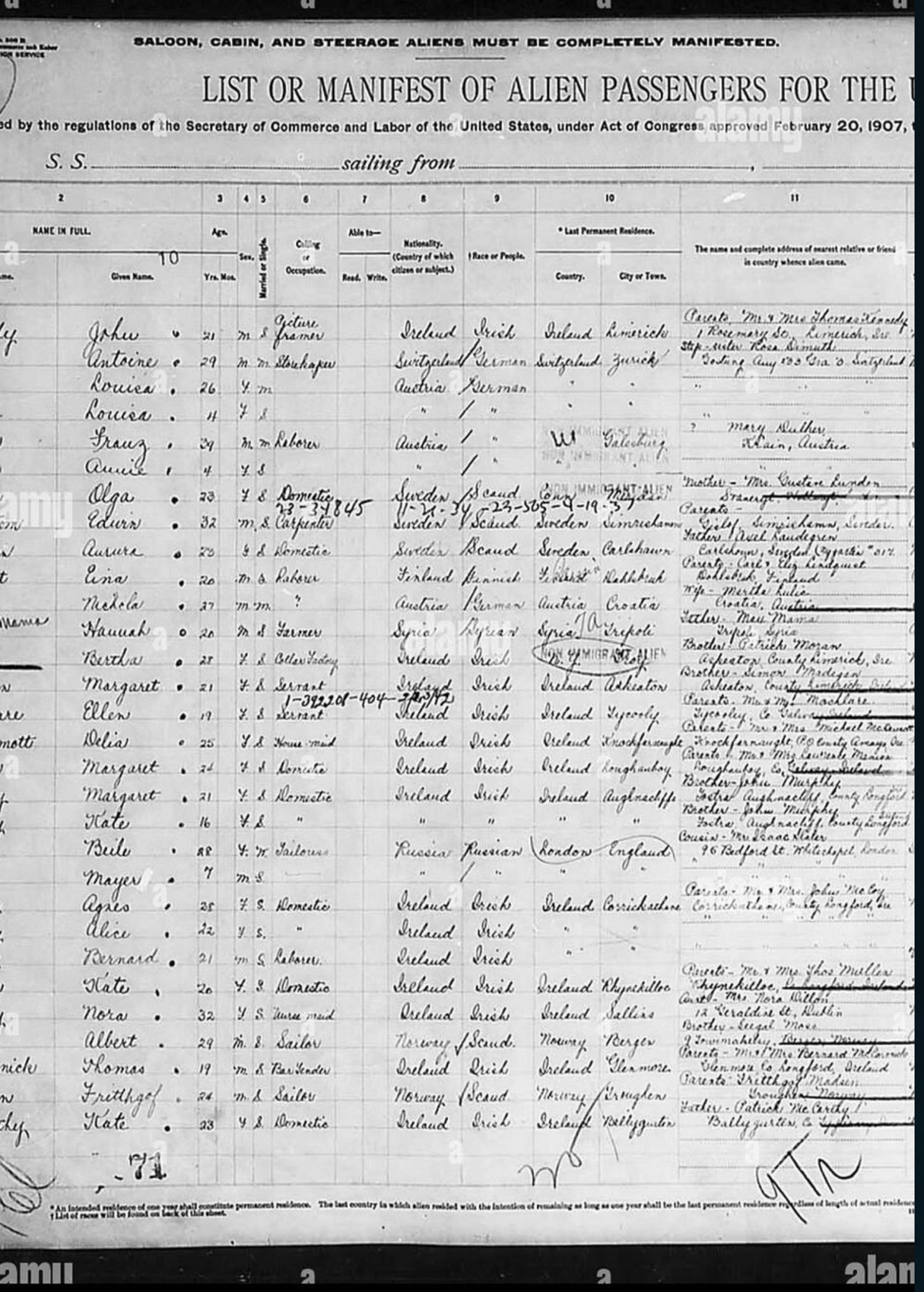


Project Objective

The Question

What factors determined who survived the Titanic disaster?

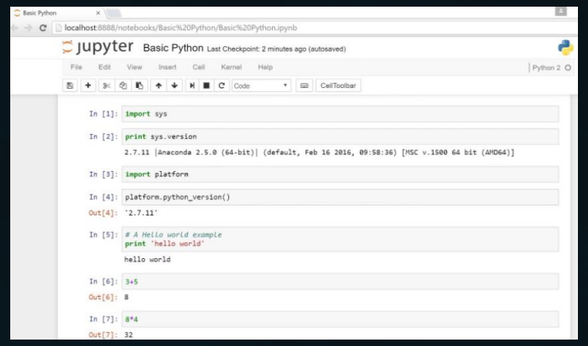
The Dataset

891 passengers · 11 variables · Real historical data

The Goal

Apply Pandas to explore, clean, and analyze real-world data like a professional analyst

Tools & Methodology



Tools Used

Python 3.x	Programming language
Pandas	Data manipulation & analysis
NumPy	Numerical operations
Jupyter Notebook	Interactive analysis

Analysis Process

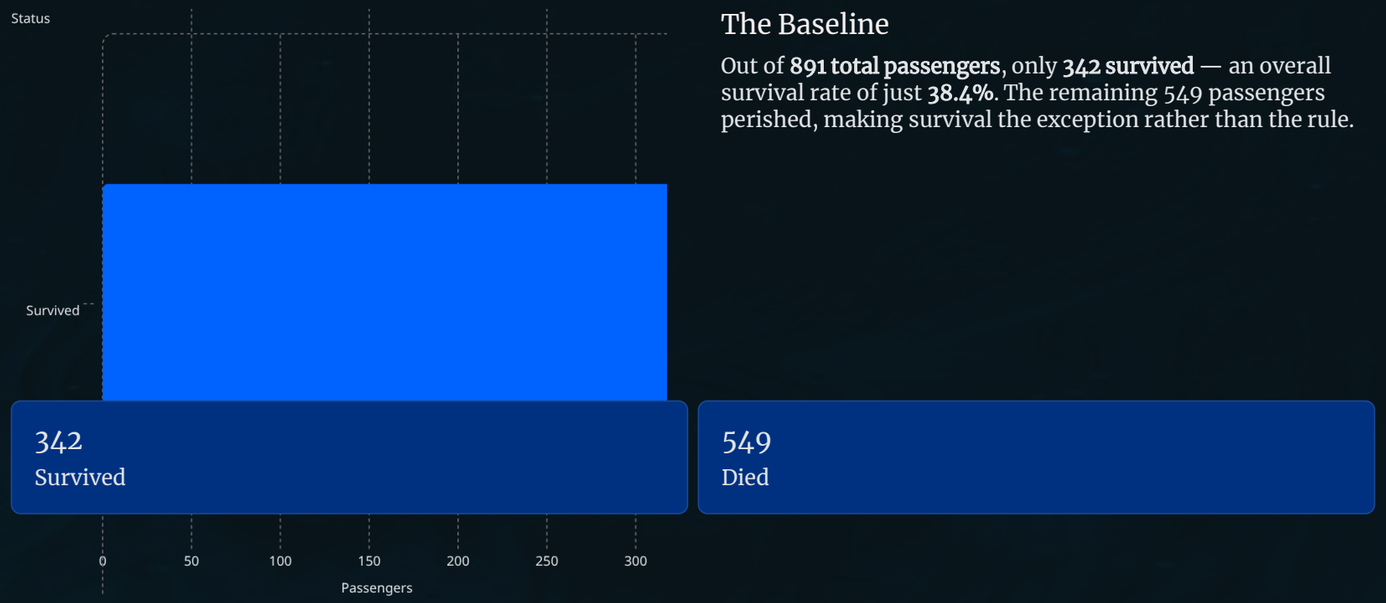
1. Load Data
2. Clean Data
3. Analyze
4. Conclude

Data Cleaning — Before & After

Issue	Count	Solution
Missing Age	177 values	Filled with median age by passenger class
Missing Cabin	687 values	Created "Has Cabin" (Yes/No) feature
Missing Embarked	2 values	Filled with most common port (Southampton)

Result: Clean, analysis-ready dataset with no missing values.

Overall Survival Rate



The Baseline

Out of **891 total passengers**, only **342 survived** — an overall survival rate of just **38.4%**. The remaining 549 passengers perished, making survival the exception rather than the rule.

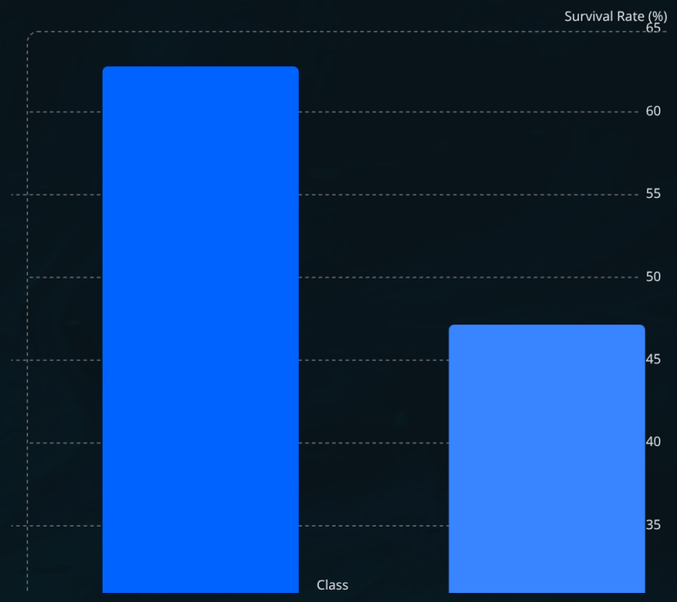
Key Finding #1 — Gender

The Numbers

Gender	Rate	Survivors	Total
Female	74.2%	233	314
Male	18.9%	109	577

Key Insight: Women were **4× more likely** to survive than men. "Women and children first" was strictly followed.

Key Finding #2 — Social Class

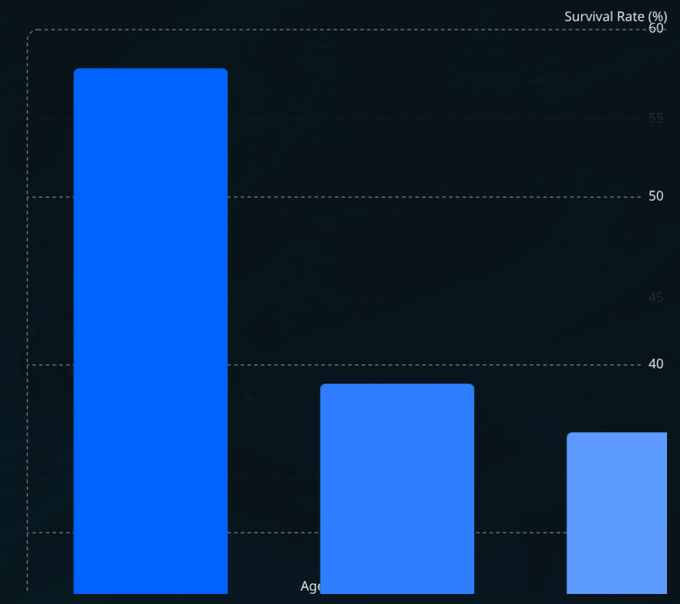


The Numbers

Class	Rate	Survivors	Total
1st Class	62.9%	136	216
2nd Class	47.3%	87	184
3rd Class	24.2%	119	491

Key Insight: 1st class passengers were **3× more likely** to survive than 3rd class.

Key Finding #3 — Age

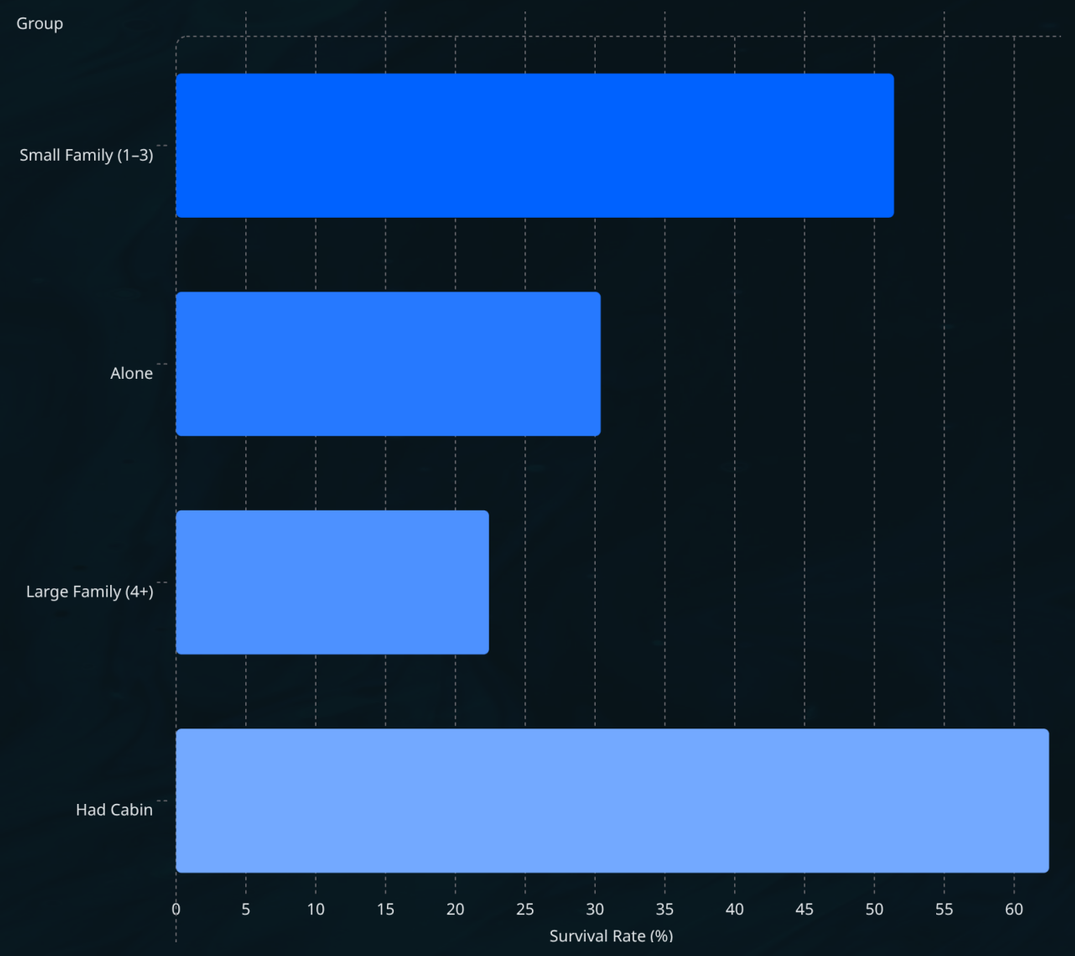


The Numbers

Age Group	Rate	Survivors	Total
Child (0–12)	57.7%	45	78
Teen (13–18)	38.9%	37	95
Adult (19–50)	36.0%	231	642
Senior (51+)	22.2%	16	72

Key Insight: Children had the **highest survival rate (58%)**, while seniors had the lowest (22%).

Additional Factors



Family Size

Category	Survival Rate
Small Family (1–3)	51.4%
Alone	30.4%
Large Family (4+)	22.4%

Cabin Ownership (Wealth)

Status	Survival Rate
Had a Cabin	62.5%
No Cabin	30.0%

Key Insight: Small families and wealthy passengers had better survival chances.